

Ice Task 6

Software Development Lifecycle (SDLC)

A structured method for designing, developing, testing and deploying software systems through predetermined stages including planning, design, implementation, testing and maintenance is known as the Software Development Lifecycle (SDLC). This systematic approach guarantees software development efficiency, quality and consistency (NIST, 2022).

Secure Software Development Lifecycle (SDLC)

This approach incorporates security procedures into each stage of the conventional SDLC. By lowering vulnerabilities and enhancing system resilience it guarantees that security is taken into account from the very beginning of development (OWASP, 2023; NIST, 2022).

DevSecOps

DevSecOps is a software development methodology that encourages cooperation between the development, security and operations teams in order to include security into DevOps procedures. To improve system security, it places a strong emphasis on automation, ongoing integration and early vulnerabilities identification (IBM, 2023; Microsoft, 2023).

Banking web applications

Online platforms known as banking web applications allow users to carry out financial transactions such account management, transfers and payments. These systems frequently become targets of cyberattacks due to financial data being sensitive requiring strong security measures (SABRIC, 2024).

Cybersecurity

The practice of defending systems, networks and data from online threats, illegal access and harm is known as cybersecurity. It entails putting procedures and technologies in place to guarantee the availability, confidentiality and integrity of data (NIST, 2022).

Vulnerabilities

Vulnerabilities are defects or weaknesses in software systems that an attacker could use to undermine security. One of the main goals of secure software development techniques is to find and fix vulnerabilities (OWASP, 2023; NIST, 2022).

References

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